

CAPSID: A Private Session ID System for Small UAVs

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1.1 unmanned aerial vehicles (UAVs)

①Application



②Threat



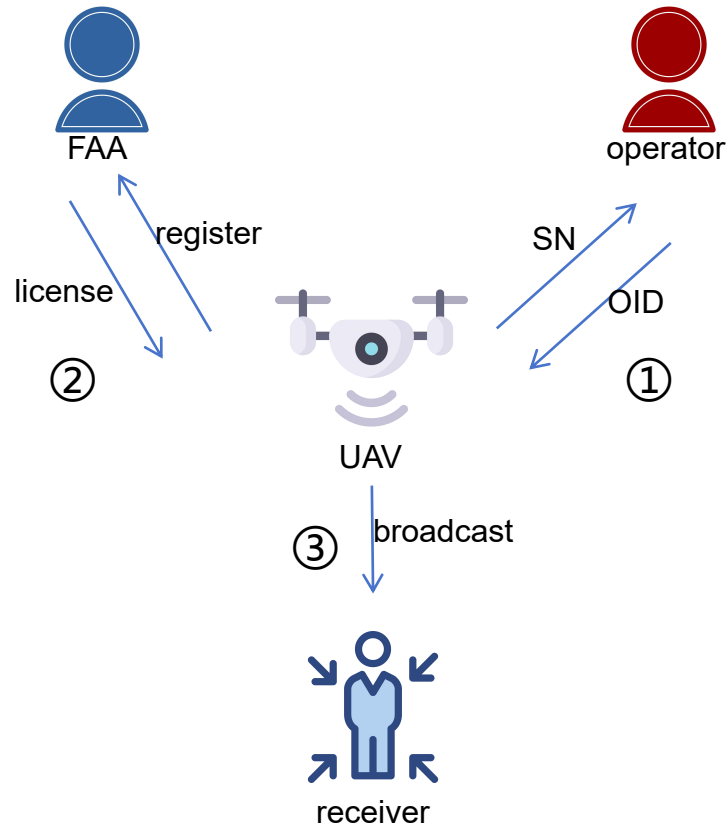
图2 2015年1月一架小型无人机坠入白宫



图3 2015年4月日本首相官邸房顶上发现无人机

1.2 BACKGROUND

Remote ID system



Broadcast messages:

- **UAV:** latitude 、 longitude、 velocity、 serial number(SN)、 session ID
- **control station:** latitude 、 longitude
- The current time (UTC)
- An indication of the emergency status of the UAV.

Problems with Remote ID

- Unverifiable by unprivileged receivers.
- Unauthenticated.
- Mass surveillance.

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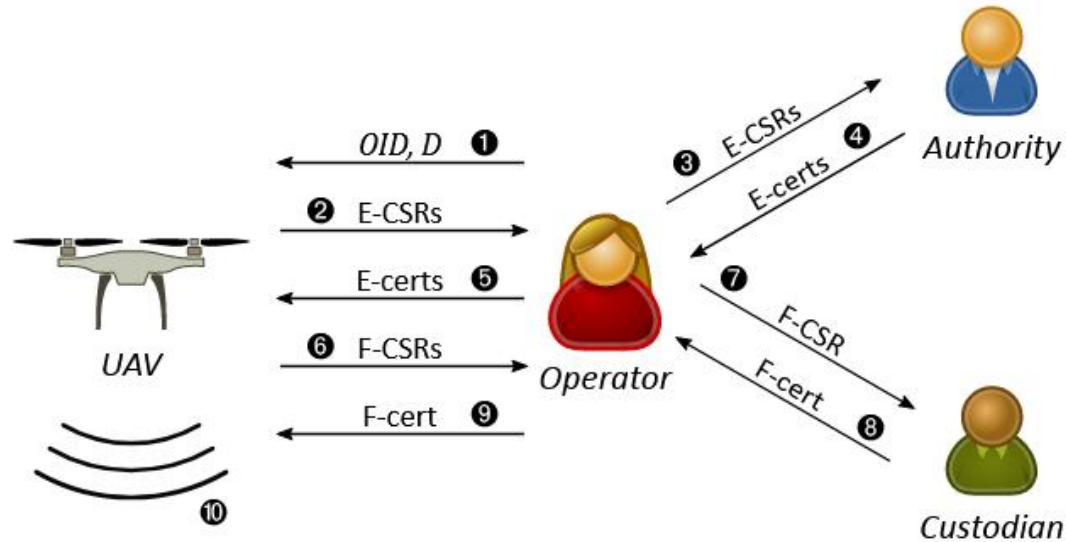
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2.1 CAPSID



- OID: Operator identifier
- D: The interval between the operating cycles
- E-CSRs: epoch certificate signing request
- E-certs: epoch certificate
- F-CSRs: flight certificate signing request
- F-certs: flight certificate

① Enrollment phase

- Operator registration: OID
- UAV registration: E-cert
- flight subscription: $F\text{-cert}(K^{UF})$

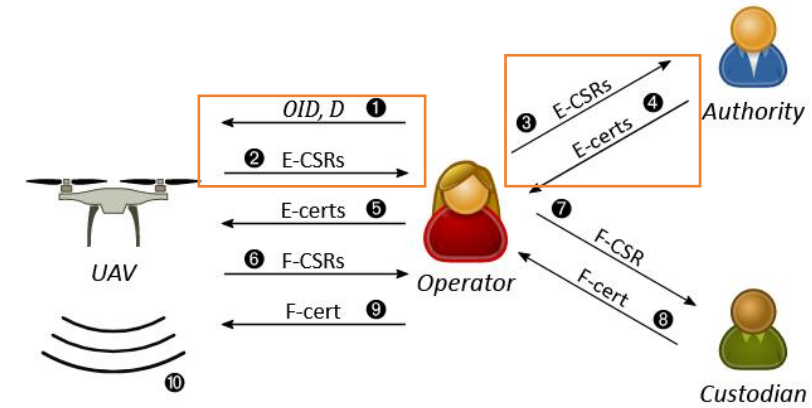
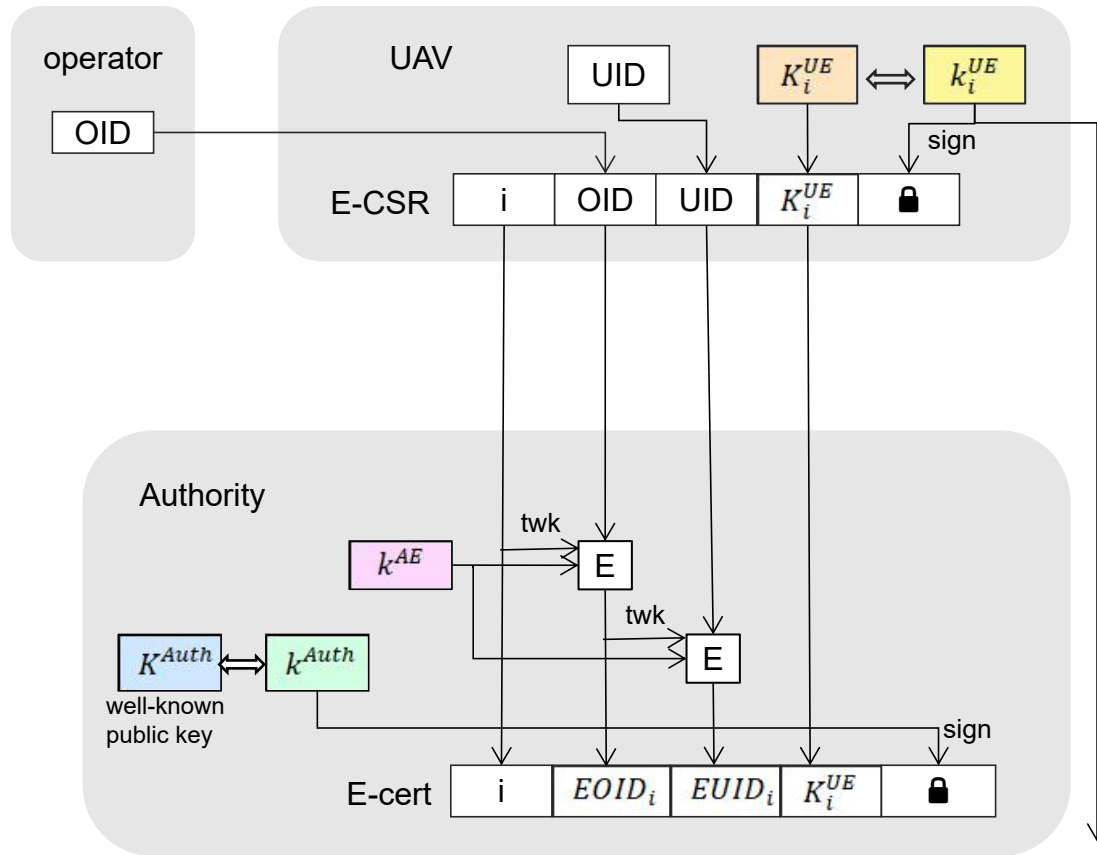
② Operational phase

- Broadcast signing: message, F-cert
- Unsealing.

◆ security properties:

- Verifiable.
- Traceable.
- Unlinkable.

2.2 UAV registration



$$E_{CSR}(i, OID, UID, K_i^{UE}, k_i^{UE}) = (i, OID, UID, K_i^{UE}, \sigma_{ecsr}),$$

$$\sigma_{ecsr} = S_{k_i^{UE}}(i \parallel OID \parallel UID \parallel K_i^{UE})$$

$$E_{CERT}(i, OID, UID, K_i^{UE}) = (i, EOID_i, EUID_i, K_i^{UE}, \sigma_{cert}),$$

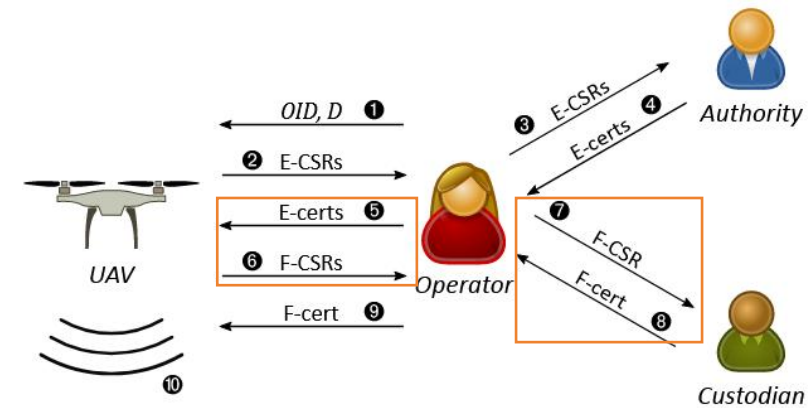
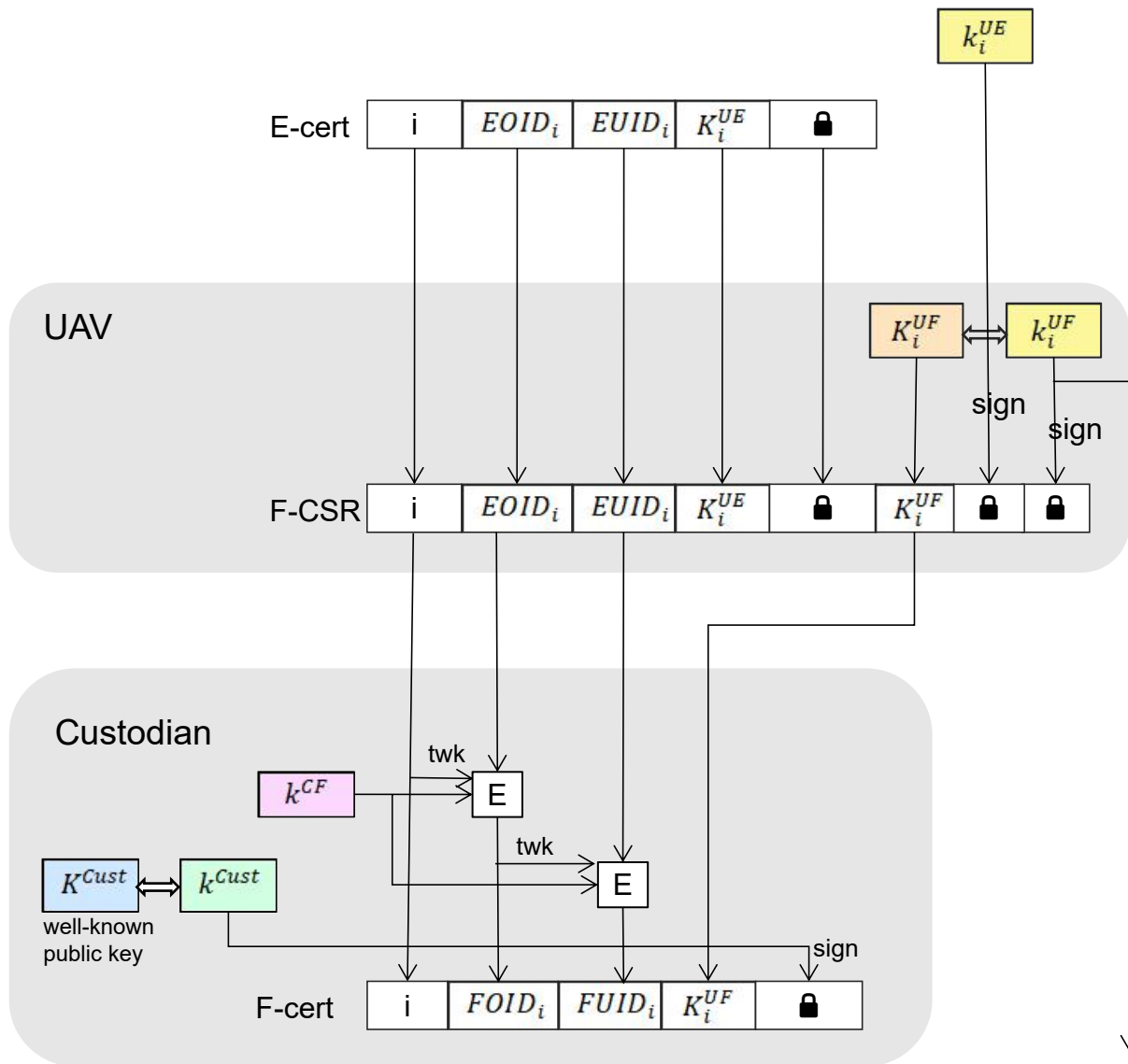
$$\sigma_{cert} = S_{K^{Auth}}(i \parallel EOID_i \parallel EUID_i \parallel K_i^{UE})$$

$$EOID_i \equiv \tilde{E}_{k^{AE}}(0 \parallel i, OID),$$

$$EUID_i \equiv \tilde{E}_{k^{AE}}(1 \parallel EOID_i, UID)$$

$\tilde{E}_k$: tweakable block cipher
i: epoch number

2.3 Flight Subscription



$$F_{CSR} \left((i, EOID_i, EUID_i, K_i^{UE}, \sigma_{ecert}), k_i^{UE}, K_i^{UF}, k_i^{UF} \right) \\ \equiv \left((i, EOID_i, EUID_i, K_i^{UE}, \sigma_{ecert}), K_i^{UF}, \sigma_{fcsr}^{UE}, \sigma_{fcsr}^{UF} \right)$$

$$\sigma_{fcsr}^{UE} = S_{k_i^{UE}}(i \parallel EOID_i \parallel EUID_i \parallel K_i^{UE} \parallel \sigma_{ecert} \parallel K_i^{UF})$$

$$\sigma_{fcsr}^{UF} = S_{k_i^{UF}}(i \parallel EOID_i \parallel EUID_i \parallel K_i^{UE} \parallel \sigma_{ecert} \parallel K_i^{UF} \parallel \sigma_{fcsr}^{UE})$$

$$F_{CERT}(i, EOID_i, EUID_i, K_i^{UF}) = (i, FOID_i, FUID_i, K_i^{UF}, \sigma_{fcert})$$

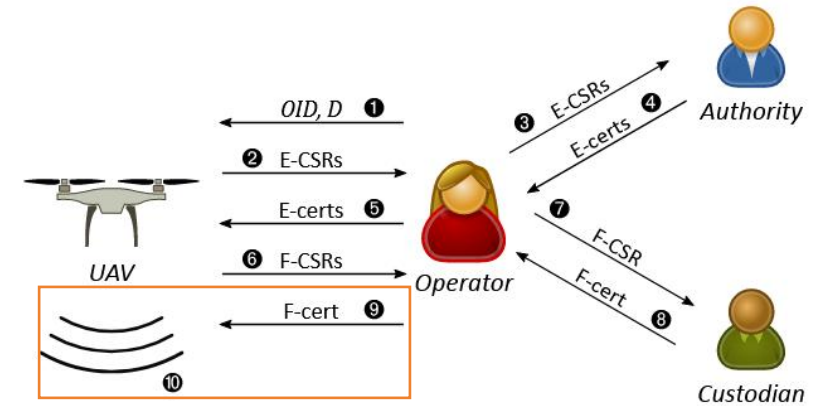
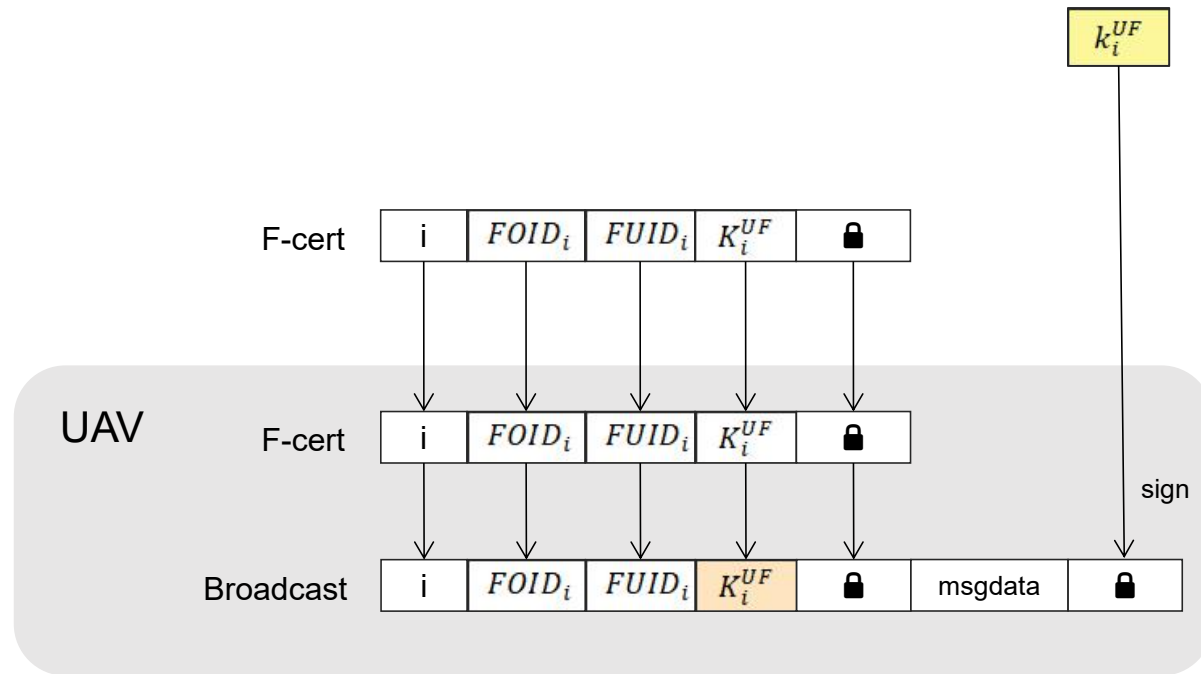
$$\sigma_{fcert} = S_{k^{Cust}}(i \parallel FOID_i \parallel FUID_i \parallel K_i^{UF})$$

$$FOID_i \equiv \tilde{E}_{k^{CF}}(0 \parallel i, EOID_i),$$

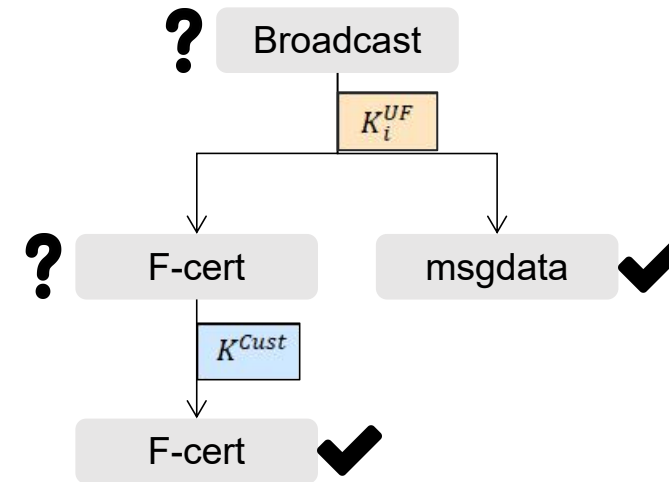
$$FUID_i \equiv \tilde{E}_{k^{CF}}(1 \parallel FOID_i, EUID_i)$$

i : epoch number

2.4 Flight Information Broadcasts



- Receiver verification:



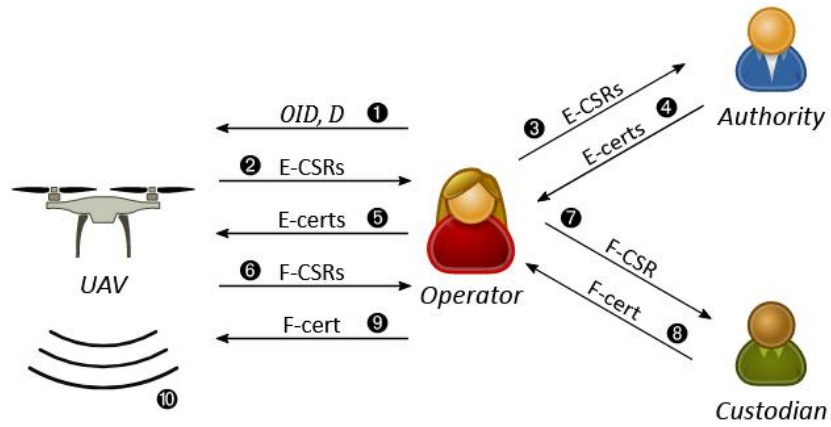


Figure 1: CAPSID protocol workflow.

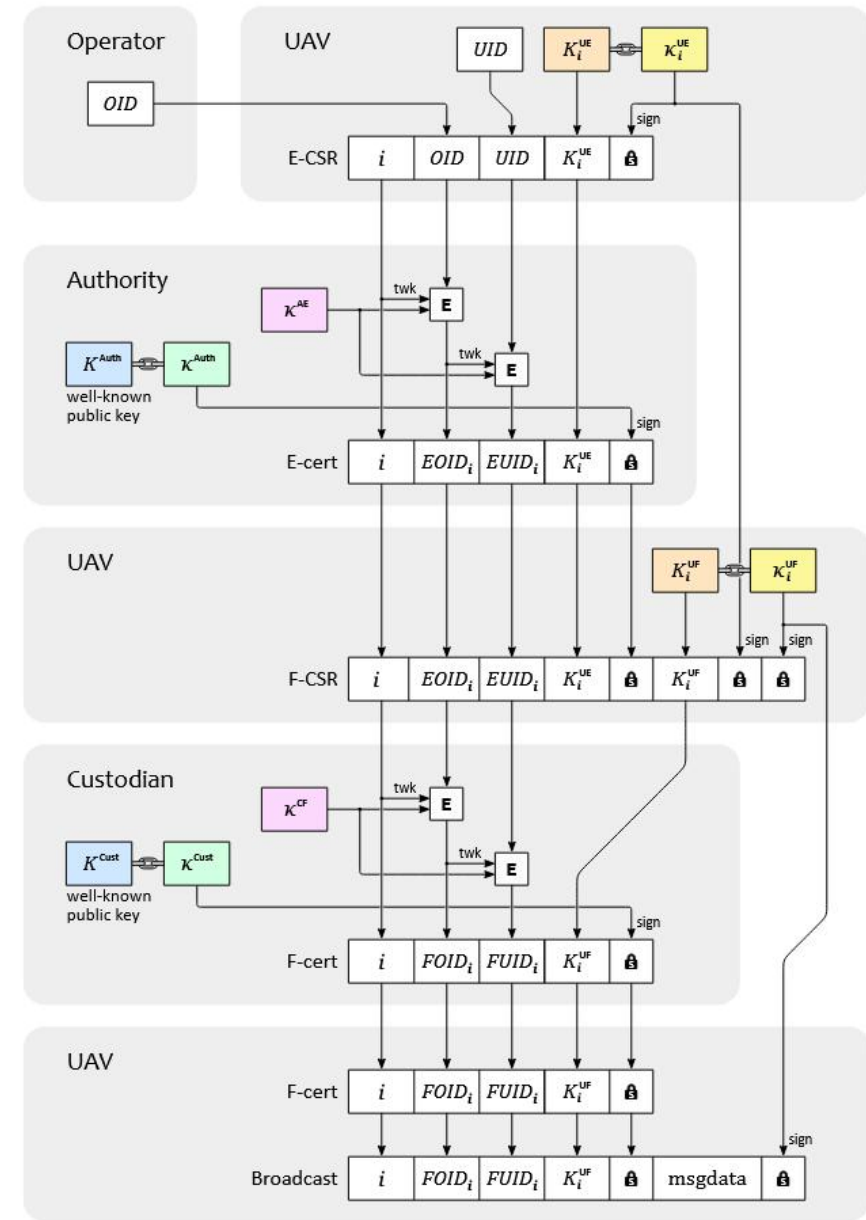
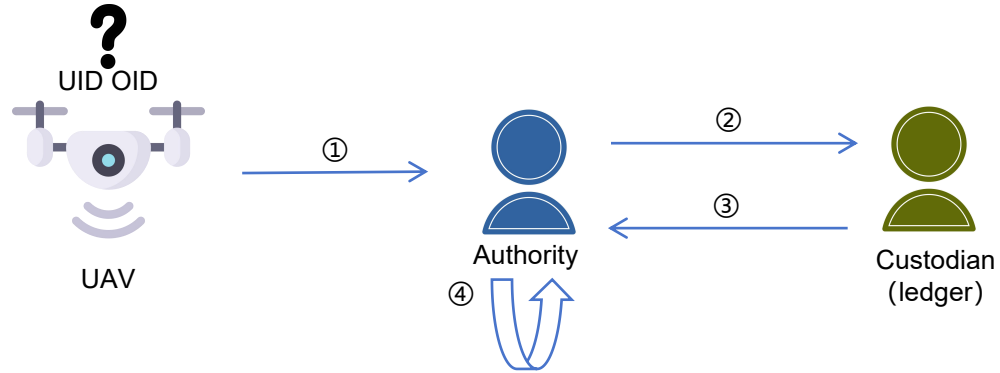


Figure 2: Certificate and broadcast construction.

2.5 Unsealing

- Unsealing process:



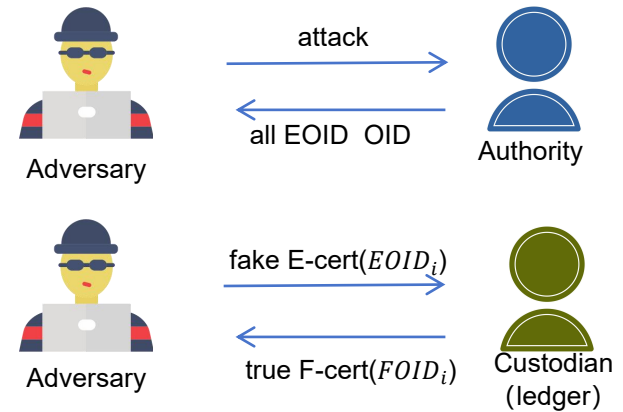
- ①: Broadcast
- ②: F-cert
- ③: EOID、EUID
- ④: OID、UID

- Custodian ledger (public, append- only)

- (1) Timestamp of the transaction,
- (2) The epoch number, EOID, and EUID from the F-CSR,
- (3) A cryptographic commitment to the the issued F-cert.

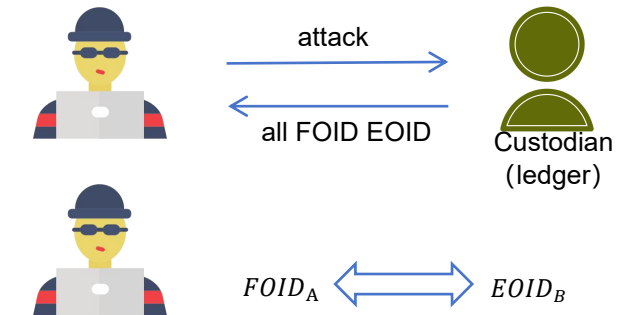
- the functions of Custodian ledger

- Corrupt Authority



Adversary can know all OID_i 's broadcast

- Corrupt Custodian



unsealing: $FOID_A$ ' broadcast $\xrightarrow{\text{traced}}$ operator(OID_B)

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3.1 Verifiability and Traceability

- Modeling Tool : *ProVerif*

Constructors:

- E-certs, F-certs: Verified via Authority keys and Custodian keys.
- Broadcasts: Encrypted with tweakable block ciphers and signed keys.
- Ledger: Immutable, append-only for attacker.

Attacker capabilities :

- Unlimited Authority and Custodian interaction.
- Full access to broadcast channel.
- Limited control of legitimate operators.
- Compromised Custodian.

- Issue three verification queries to ProVerif, described below.

Query 1: Verifiable broadcasts.

- Assertion:
 - if:
 - *a broadcast is accepted by a receiver*
 - then:
 - $F_{CERT} \leftarrow \text{Custodian}$
 - $E_{CERT}(i, EOID, EUID) \leftarrow \text{Authority}$
- Result:
 - ✓ Passes in Honest Custodian Model.
 - ✗ Fails in Corrupt Custodian Model.

Query 2: Traceable broadcasts

- Assertion:
 - pre:
 - create operator O : attacker know O's OID
 - if:
 - *a broadcast is accepted by a receiver*
 - then:
 - *message* $\overleftarrow{\text{send}}$ operator(OID)
- Result:
 - ✓ Passes in Honest Custodian Model.
 - ✗ Fails in Corrupt Custodian Model.

Query 3: Impersonation-resistant broadcasts

- Assertion:
 - if:
 - a broadcast is traced to an operator
 - the operator has accepted the F-cert
 - then:
 - *message* $\overleftarrow{\text{send}}$ operator(OID)
- Result:
 - ✓ Passes in Honest Custodian Model.
 - ✗ Fails in Corrupt Custodian Model.

3.2 Operator Non-linkability

- E-certs and F-certs construction:

$$E_{CERT}(i, OID, UID, K_i^{UE}) = (i, EOID_i, EUID_i, K_i^{UE}, \sigma_{cert}),$$

$$F_{CERT}(i, EOID_i, EUID_i, K_i^{UF}) = (i, FOID_i, FUID_i, K_i^{UF}, \sigma_{fcert})$$



- common construction:

$$P_{CERT_k}(i, o, u) = (\tilde{E}_k(0 \parallel i, o), \tilde{E}_k(1 \parallel \tilde{E}_k(0 \parallel i, o), u))$$

$$\tilde{E}_k: \{0,1\}^k \times \{0,1\}^\ell \times \{0,1\}^n \rightarrow \{0,1\}^n$$

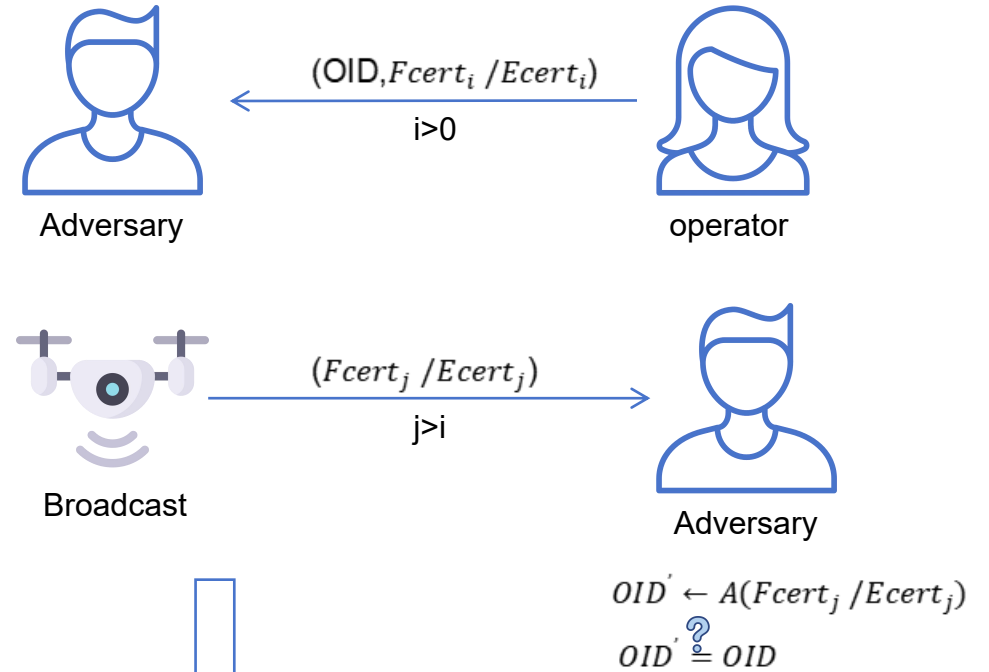
$$\ell = n+1$$



proof: Non-linkability



proof: Indistinguishability



3.3 Non-Linkability Experiment

- *left-or-right function:*

$$LR(x, y, 0) = x \quad , \quad LR(x, y, 1) = y$$

- *Pcert oracle:*

$$O_{k,b}(i, o_0, u_0, o_1, u_1) = P_{CERT_k}(i, LR(o_0, o_1, b), LR(u_0, u_1, b))$$

Oracle $O_{k,b}(i, o_0, u_0, o_1, u_1)$ is

- ① if $o_0 \neq o_1$ or $u_0 \neq u_1$ then
 - if $(i, o_0) \in Q_R \cup Q_C$ or $(i, o_1) \in Q_R \cup Q_C$ then
 - $fair \leftarrow FALSE$
 - end
 - $Q_C \leftarrow Q_C \cup \{(i, o_0), (i, o_1)\}$
 - ② else
 - if $(i, o_0) \in Q_C$ then
 - $fair \leftarrow FALSE$
 - end
 - $Q_R \leftarrow Q_R \cup \{(i, o_0)\}$
- end
- return $PCERT_k(i, LR(o_0, o_1, b), LR(u_0, u_1, b))$

- require(for a given epoch i , OID) :

① challenge queries: once

② regular queries: any number of times

Q_C : A collection of challenge queries

Q_R : A collection of regular queries

Experiment $\text{Exp}_{\tilde{E}}^{nlmk-b}(A)$ is

$fair \leftarrow TRUE$

$Q_R \leftarrow \emptyset$

$Q_C \leftarrow \emptyset$

$\kappa \xleftarrow{R} \{0, 1\}^k$

$c \leftarrow A^{O_{\kappa,b}}()$

if $fair$ then

return c

else

return $1 - b$

$$\text{Adv}_{\tilde{E}}^{nlmk}(A) = \Pr[\text{Exp}_{\tilde{E}}^{nlmk-1}(A) = 1] - \Pr[\text{Exp}_{\tilde{E}}^{nlmk-0}(A) = 1]$$

$$\text{Adv}_{\tilde{E}}^{nlmk}(k, n, t, q) = \max_A \{\text{Adv}_{\tilde{E},A}^{nlmk}\} \stackrel{?}{<} \text{negl}$$

k : keys size ,

n : block size ,

t : running time ,

q : number of queries

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4.1 Discussion

- Candidates for the Role of Custodian

- Another government agency.



- A UAV pilot or industry association.



- UAV manufacturers.



- Relate Work

System/Protocol	Domain	Trace bility	Verifiabili ty	Non-linkability
CAPSID	UAV	✓	✓	✓
Remote ID	UAV	✗	✗	✓
ADS-B	Manned aircraft	✓	✗	✗
VANETs	Vehicular Networks	✓	✓	✓
Group Signatures	Distributed environment	✓	✓	✓
Identity Escrow	Identity management	✓	✓	✓